

A Transfer Learning-Based Deep CNN Approach for Classification and Diagnosis of Acute Lymphocytic Leukemia Cells

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Abstract— Leukemia is a severe malignant illness affecting both adults and children. It can be chronic or acute. Acute Lymphoblastic Leukemia (ALL) is the most prevalent type of blood disease and one of the leading causes of death. This type of cancer invades the blood and spread through neighboring organs and body systems. To classify cancer cells and non-cancer cells, specialists need to perform manual diagnosis through inspection of cell images under a microscope and provide labels through annotation. However, this manual microscopic analysis is tedious and may lead to a false diagnosis. With this concern, the author proposed a computer-aided diagnosis method using a transfer learning-based deep learning approach. In this study, the YOLOv3 model is utilized to train a deep learning model that will classify ALL and normal cells. The model produced promising results as it has a training loss of 2.8% or a 97.2% training accuracy. Based on the model evaluation, the model has an mAP value of 99.8% as well. YOLOv3 was shown to be a useful tool for distinguishing leukemia cells from non-leukemia cells.

Keywords— leukemia cell classification, deep learning, transfer learning, YOLOv3, cancer cell detection

I. INTRODUCTION

Cancer is one of the world's deadly diseases. It is a general term covering a wide range of illnesses that can affect any system or aspect of the human body [1]. The most common kind of blood cancer in persons of all ages is leukemia. Acute lymphocytic leukemia (ALL) and Acute myeloid leukemia (AML) are the two types of malignancy. Because of excessive blood cell proliferation and expansion, this abnormality can affect the body's defensive system, as well as the red blood cells [2]. Furthermore, these abnormal leukemia cells can rapidly move into the bloodstream and travel to other organs such as the liver, spleen, kidneys, lymph nodes, and brain, causing a variety of cancers [3]. Leukemia is classified as lymphocytic or myelogenous depending on the type of white blood cell affected. When the infantile cells are granulocytes and monocytes, the illness is defined as myelogenous, and when the infantile cells are lymphocytes, the illness is categorized as lymphocytic [3].

Acute Lymphoblastic Leukemia is the subject of this research report. Among pediatric cancers, it is the most prevalent kind of malignancy that develops in white blood cells in the bone marrow (soft inner portion of bones) and accounts for around 25% of all cases in children. Children have a greater recovery percentage than elders [4]. From an immature lymphocyte, a type of essential white blood cell inside the immune system, this type of cancer develops. It was once thought to be an uncommon kind of leukemia since it grows worse fast. It affects both adults and children, however, it is more frequent in children [5].

Researchers, clinicians, and hematologists have often been challenged to make an early diagnosis of leukemia. A complete platelet count is frequently used to assess acute lymphocytic leukemia [6]. During this test, the doctor will look to see whether the number of WBCs grows and if there are any indications of leukemia cells. And, in other cases, these indications are insufficient for a specialist to declare leukemia in a patient. [3]. As a result, they must use a range of techniques, such as bone marrow aspiration and blood smear microscopic inspection, and the majority of these manual operations require a lot of time and effort [7]. Furthermore, skilled and experienced medical personnel are necessary to undertake these sorts of examinations, making it a time-consuming process [8]. Meanwhile, automated diagnostic technologies have the potential to overcome the shortcomings of manual diagnosis. Also, as compared to manual diagnosis, it will reduce healthcare practitioners' stress while providing efficient and precise results [3].

To overcome these constraints, studies have offered several computer-aided diagnosis approaches for all instances when microscopic blood image analysis is used to identify leukemia. These procedures were found to be more efficient, quick, accurate, and cost-effective when compared to conventional approaches. [6].

In the study of Lim Huey Nee et al, they suggested techniques for white cell segmentation according to the feature extraction process, gradient intensity, and watershed transform [9]. Following the collection of image data, segmentation is performed to distinguish the blast cell from

the background. For this process, the RGB picture is first converted to an HSV color model, and the saturated element is eliminated for further processing before calculating the gradient magnitude for the saturated component [9]. This is used to detect edges. On the other hand, in the work of Moradiamin et al, they converted RGB pictures to HSV, which lessened the coherence between the color channels contrasted to RGB and allowed the three H, S, and V channels to be treated individually. [10].

For the study, Thanh et al created a CNN model with five convolutional layers to produce a binary classification of ALL leukemia subtypes and achieved 96.6% accuracy [11]. Unfortunately, the performance classification in this type of neural network requires a huge amount of training data to learn how to recognize key features from the entire image. Creating a huge training dataset, on the other hand, takes a long time and is a labor-intensive process [11]. To circumvent this issue, they propose increasing the limited number of samples through image augmentation. Another is the study of Purwanti et al, in which they used a combination of shape and histogram features and a k-nearest neighbor classification algorithm.

Some of the suggested or proposed algorithms use image pre-processing sequentially, cell segmentation, image segmentation or extraction, and cell categorization and evaluation. Picture enhancements such as contrast adjustment, noise reduction, and image sharpening are included in the pre-processing stage to eliminate microscopic noises to achieve high detection accuracy. These datasets, in different studies, are trained using different algorithms, such as SVM, KNN, CNN, etc. This study utilized these preprocess datasets along with a pretrained algorithm YOLOv3.

This study also proposed a transfer-learning-based deep learning approach for Acute Lymphoblastic Leukemia classification. The YOLOv3 method will be used as it can provide excellent results in terms of accuracy and minimal complexity [12]. Its structure will be discussed as well as the metric values used to evaluate the trained model. In the next sections, methodology, results, and discussions as well as the conclusion will be covered and discussed.

II. METHODOLOGY

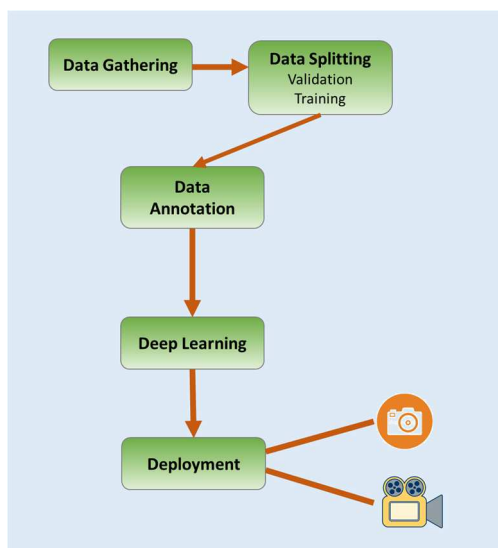


Fig. 1. Conceptual Framework

In this section, the steps done in the research are discussed thoroughly. Each stage will be well-explained as well as the tools utilized. These steps include data gathering and dataset preparation, image annotation, model training and evaluation, and deployment. This is further demonstrated in Fig. 1.

A. Data Gathering and Dataset Preparation

The dataset utilized in this study was obtained from Kaggle.com, a popular data science website. [13]. It was comprised of cells segmented from microscopic images and is a representation of real-world images. It has some discoloration noise and luminescence errors, but these have been mostly corrected during the acquisition phase.

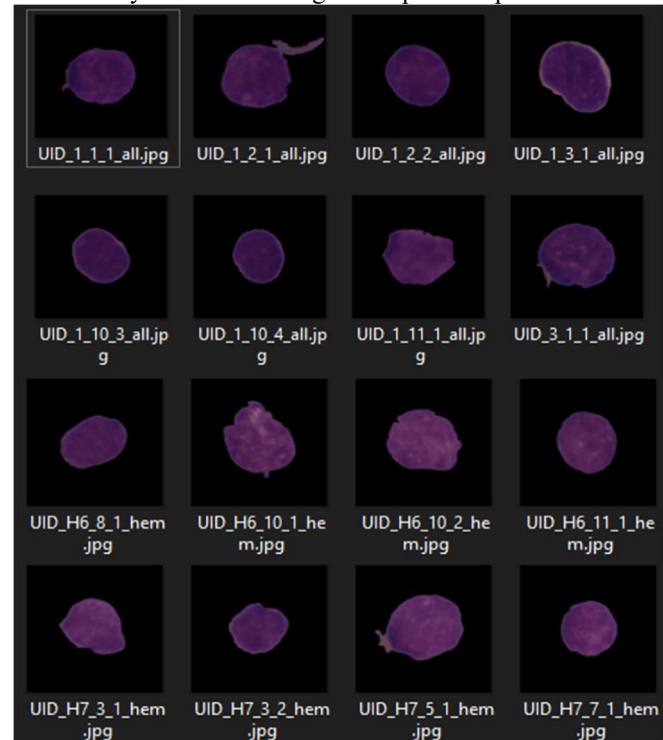


Fig. 2. Image Samples from Dataset

There are 15,135 images in total from 118 patients, with two labeled classes: Normal Cell and Leukemia Blast. Only 300 from these images are selected to be used for model training and evaluation. Seventy percent (70%) or 240 of these images are used for training, with the remaining thirty percent (30%) or 60 images being used for validation. Since there are two classes or labels, the selection of data must be equal. In this case, 150 are from (HEM) Normal Cell, and the other half are from Leukemia Blast (ALL). The training images consist of 120 HEM-labelled (Hemocyte) images and another 120 ALL-labelled (Acute Lymphoblastic Leukemia) images. On the other hand, the validation images consist of 30 HEM-labelled images and another 30 ALL-labelled images. Shown in Fig. 2 are sample of dataset images used in the study.

B. Image Annotation

Image annotation is the task of annotating images with labels. Usually, it involves manpower, and, in some cases, computer-aided [8]. The labels are predetermined already and are chosen to provide the deep learning model the information it needs as shown in Fig. 3. This process also

helps the AI model to perform well and yields good results in terms of overall precision and accuracy.

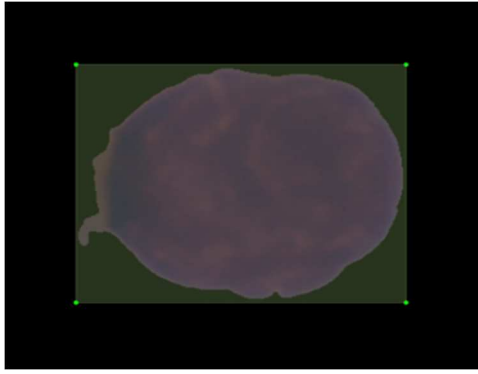


Fig. 3. Annotated Cell Image

C. Model Training and Evaluation

For model training, the YOLOv3 deep learning algorithm is used by the researcher. The ‘You Only Look Once’ v3 or YOLOv3 is among the most extensively used deep learning-based object detection methods. It predicts bounding boxes in the entire image, which are then divided into multiple parts using a single convolutional neural network (CNN). The CNN becomes more effective because of its capacity to make several predictions at once.

The researcher used YOLOv3, given its framework and architecture in Fig. 4. It is easy to use with its fundamental structure and wide usage. Usually, it is suggested to perform a data augmentation procedure to improve the dataset as well as the performance of the model. Filtering, cropping, HSV distortion, and flipping are just a few of the techniques that may be used. In this study, the author did not undergo data augmentation as the given dataset has enough samples and was preprocessed already. Compared with other algorithms like SSD and Faster R-CNN, in terms of mean average precision (mAP) and intersection over union (IOU) values, YOLOv3 is also quick and precise. It is substantially faster than other related detection methods [14]-[22].

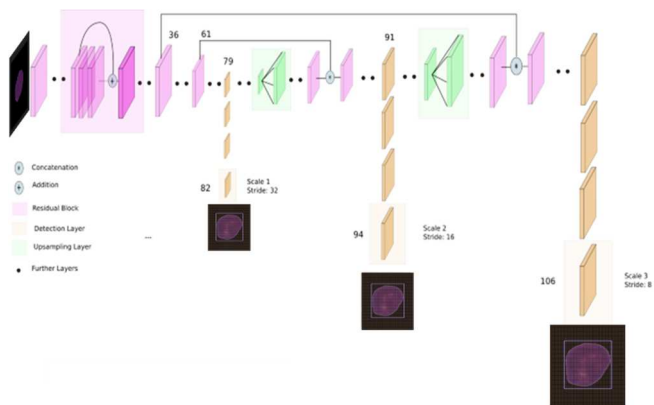


Fig. 4. YOLOv3 Framework

Google Colab was used to train the model. It is a Google Research product that allows anyone to write and execute Python code while hosting Jupyter notebooks in their browser. It is especially well-suited for machine learning and data analysis. It also grants free access to computing resources such as graphical processing units.

After the training phase, the most well-trained model was picked for deployment. The trained models are evaluated using mean Average Precision (mAP) (1) to determine the best-performing model.

$$mAP = \frac{\sum_{i=1}^o AP_i}{o} \quad (1)$$

The given formula in (1) calculates the corresponding AP for a given query o , and the mean of all those AP scores is the mAP. It quantifies the trained model's performance. Also, during the dataset training and model evaluation, configurations such as batch sized=4, you (intersection over union) threshold=0.5, object threshold=0.3, non -maximum suppression threshold=0.5 are used to fine-tune the model in transfer learning.

D. Deployment

The Anaconda IDE, PyQt5, and the ImageAI detection module are all used to develop a graphical user interface (GUI) in this study. It has three main functions: image detection based on image file inputs, video detection based on the video file input variables, and live feed detection based on the camera device on a computer. It makes use of the JSON configuration file and the specified deep learning model h5 file.

New data which are not included in the training dataset must be used to ensure that there will be no biases that may affect the accuracy (2) of the model. For live feed detection, the cell image was shown using printed materials as well as for video detection.

$$accuracy = \frac{\# \text{ of Correctly Detected Images}}{\text{Total \# of Images}} \times 100 \quad (2)$$

III. RESULTS AND DISCUSSIONS

The training, validation, testing, and deployment results will be discussed in this section.

A. Modeling Results

The outcomes of the training and evaluation are depicted in Fig. 5. The number of epochs to be gained is set to 25 at the start, and the training level is set to 25 at the start. However, during training, some model (h5) files are missing, hence the author employed additional training levels to acquire a total of 25 epochs, resulting in a total of 32 training levels.

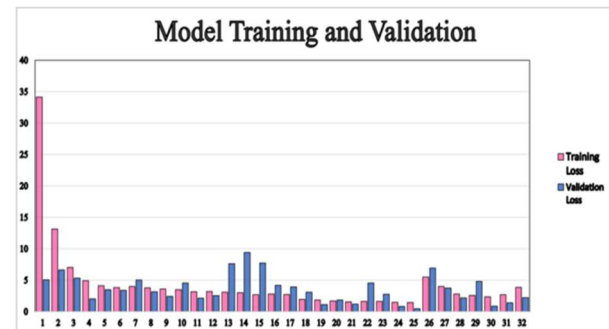


Fig. 5. YOLOv3 Model Training and Validation Result

During the initial training stage, it yields a train loss (represented by pink bars) of approximately 34.14% and a validation loss (represented by blue bars) of approximately 5.07%. However, at the last training stage, it had a train loss of only 3.87% and a validation loss of 2.22%. As shown in Fig. 6, at the 25th training stage, it yields the lowest train and validation loss values of 1.44% and 0.47% correspondingly. On the other hand, it was the 1st training stage that yields the highest train and validation loss values.

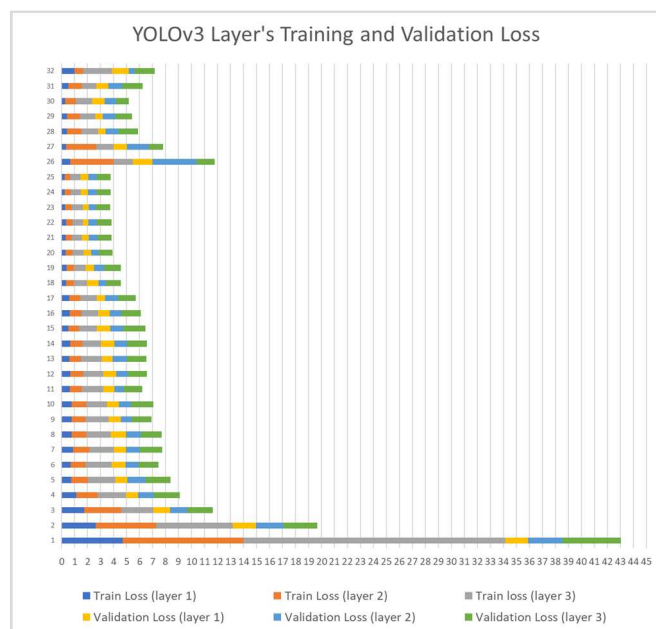


Fig. 6. YOLOv3 layers' training results

It was also concluded, as shown in Table 1 that the average runtime is 607.69 frames per seconds, the average train loss is 4.37%, and the average validation loss is 3.65%.

TABLE I. YOLOV3’S TRAINING RUNTIME

Averages	
Runtime	607.69 frame per sec
Train loss	4.37%
Validation loss	3.65%

B. Evaluation Findings

In terms of assessment, the mAP indicates the dataset's precision. mAP equals 1 or 100 percent [1], and the acceptability value is near to the highest possible value because it is expressed in percentages. The mAP of every model is shown in Fig. 7.

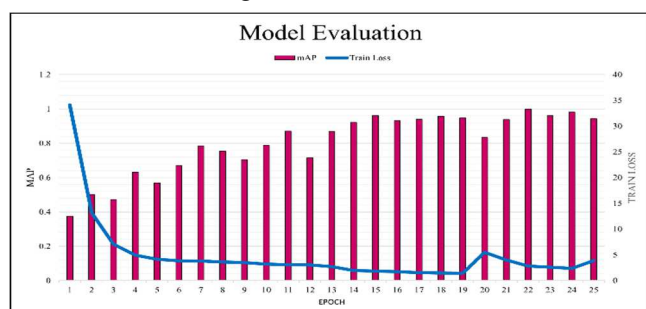


Fig. 7. Evaluation Results

It was concluded that epoch no. 22 obtained an mAP value of 0.998 (99.8%) and was considered the highest performing model. Its training loss value is 2.8% and the validation loss is 2.18% as shown in Fig. 7 (no. 28).

C. Deployment Findings

The created simple GUI design is shown in Fig. 8. It has three main functions: To get an image file input, a video file input, and to use the camera device. The results are stored in a CSV file.

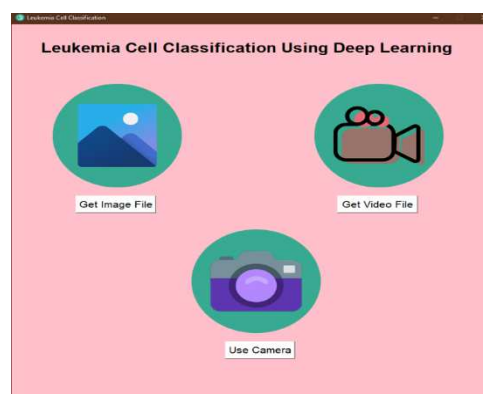


Fig. 8. Graphical User Inter DesignTesting Results

To test the trained deep learning model, the researcher used to image and video file inputs and a camera device for live feed detection. The live feed output is shown in Fig. 9, and it is evident that the detection accuracy is changing or varying over time. For cell image detection, the researcher used printed materials that will serve as the test input for the deployment. The images used were fresh and not included in the training dataset. However, its ground truths were pre-determined. Fig. 10, it shows that the output for an ALL-labelled image was 89.873 and 54.947 for a HEM-labelled image.

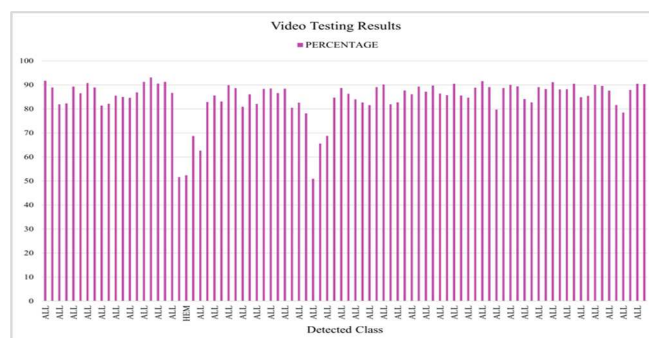


Fig. 9. Video Testing Results

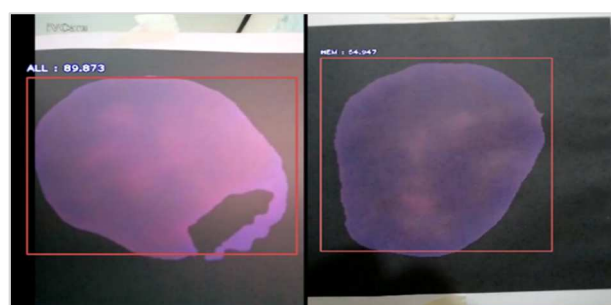


Fig. 10. Video Frames Testing Result

IV. CONCLUSION AND FUTURE WORKS

With a small number of datasets, the proponent was able to train deep learning models using the YOLOv3 algorithm and utilize CNN as well. The trained models were assessed using a popular evaluation metric for object detection - mAP. The best model was selected yielding an mAP score of 0.998. During model inference or deployment, the selected model performed and yielded promising accuracy results. It was then concluded that the YOLOv3 is a good method for identifying leukemia cells from normal cells and might be a big help in oncology medicine.

The study has certain limitations, such as the limited number of experimental data, as the collected dataset is only for Acute Lymphocytic Leukemia. It is recommended that in the future study, a new dataset can be used, for the detection of other types of leukemia or blood cancer and its subtypes.

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